

Patient Healthcare Analytics Project

This project demonstrates an end-to-end healthcare analytics workflow using Python, SQL, and Power BI. The main objective of the project is to analyze patient healthcare data, generate meaningful business insights, and visualize healthcare KPIs through an interactive dashboard.

1. Project Overview

The healthcare industry generates large amounts of patient and hospital data every day. This project focuses on analyzing healthcare records including patient information, doctor details, hospital visits, treatments, and laboratory test results. The project helps in understanding patient demographics, treatment costs, abnormal test trends, doctor workload, and healthcare performance indicators. The final output includes SQL analysis and a Power BI dashboard for decision-making.

2. Business Problem

Hospitals and healthcare organizations often struggle to manage large datasets efficiently. Without proper analysis, it becomes difficult to identify treatment trends, monitor patient outcomes, track abnormal lab tests, and optimize healthcare operations. This project aims to solve these challenges by: Analyzing healthcare data using SQL. Generating actionable KPIs. Visualizing healthcare performance in Power BI. Improving decision-making using data insights.

3. Dataset Summary

The project dataset contains healthcare-related information divided into multiple relational tables. **Tables Used:** Patient Table Doctor Table Visit Table Treatment Table Lab Test Table **Important Features:** Patient demographics (Age, Gender, Blood Type, State) Insurance information Doctor specialties and experience Visit records and diagnoses Treatment cost and outcomes Lab test results and abnormalities

4. Tools & Technologies Used

Python – Data Cleaning & Preprocessing **Pandas & NumPy** – Data Manipulation **SQL** – Data Querying & KPI Analysis **Power BI** – Dashboard & Visualization **Jupyter Notebook** – Analysis Environment **GitHub** – Project Hosting & Portfolio

5. Data Cleaning & Preprocessing using Python

The raw dataset was cleaned using Python before performing SQL analysis. **Data Cleaning Steps:** Imported datasets using pandas. Checked dataset structure using `.info()` and `.describe()`. Removed duplicate records. Handled missing values. Standardized column names. Converted incorrect data types. Prepared clean datasets for SQL

analysis. **Python Libraries Used:**

6. SQL Analysis

SQL queries were written to analyze healthcare trends and generate business KPIs. **Major SQL Operations Used:** Aggregate Functions GROUP BY CASE WHEN Subqueries ORDER BY JOIN Operations

7. Patient Analysis

The patient dataset was analyzed to understand demographics and healthcare distribution. **Analysis Performed:** Gender Distribution Age Analysis Blood Type Distribution Top States by Patient Count Insurance Provider Analysis Chronic Condition Analysis

8. Visit Analysis

The visit records were analyzed to identify patient visit trends and healthcare utilization. **Analysis Performed:** Visits by Type Visit Status Breakdown Most Common Diagnoses Reasons for Visits Follow-Up Rate Analysis

9. Treatment Analysis

Treatment data was analyzed to evaluate treatment costs and healthcare performance. **Analysis Performed:** Treatment Type Distribution Average Treatment Cost Most Expensive Treatments Treatment Outcomes Treatment Status Breakdown

10. Lab Test Analysis

Lab test data was analyzed to monitor abnormal test patterns and patient health indicators. **Analysis Performed:** Abnormal Test Result Rate Most Common Lab Tests Reference Range Analysis Abnormal Results by Test Name

11. Power BI Dashboard

An interactive Power BI dashboard was created to visualize healthcare KPIs and trends. **Dashboard Features:** KPI Cards Patient Demographics Dashboard Treatment Cost Analysis Visit Trend Analysis Lab Test Result Analysis Interactive Filters & Slicers Doctor Performance Insights

12. Key Insights

Identified common chronic conditions among patients. Analyzed treatment costs and expensive procedures. Measured follow-up rates for patient visits. Evaluated abnormal lab test percentages. Analyzed patient demographics and healthcare trends. Studied doctor workload distribution.

13. Business Recommendations

Improve patient follow-up systems for better healthcare management. Monitor abnormal lab tests for early disease detection. Optimize healthcare resources and treatment costs. Use dashboard insights for hospital decision-making. Improve preventive healthcare programs.

14. Skills Demonstrated

Data Cleaning Exploratory Data Analysis (EDA) SQL Query Writing KPI Development Dashboard Development Data Visualization Business Insight Generation Healthcare Data Analysis

15. Conclusion

This project demonstrates a complete healthcare analytics workflow from data cleaning to dashboard creation. It showcases practical skills in Python, SQL, and Power BI while generating meaningful healthcare insights that can support business and operational decisions in the healthcare industry.

Key Performance Indicators (KPIs)

KPI	Description
Total Patients	Number of registered patients
Total Doctors	Number of available doctors
Total Visits	Total hospital visits
Total Treatments	Number of treatments provided
Total Lab Tests	Number of lab tests conducted
Average Patient Age	Average age of patients
Follow-Up Rate	Percentage of visits requiring follow-up
Average Treatment Cost	Average treatment expense
Abnormal Test Rate	Percentage of abnormal lab test results
Doctor Workload	Average visits handled per doctor

Example SQL Query

```
SELECT
    Gender,
    COUNT(*) AS Patient_Count,
    ROUND(100.0 * COUNT(*) / (SELECT COUNT(*) FROM patient), 1) AS Percentage
FROM patient
GROUP BY Gender
ORDER BY Patient_Count DESC;
```

Example Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('patient_data.csv')

df.info()
df.describe()

df.drop_duplicates(inplace=True)
df.fillna(0, inplace=True)
```

Author: Bhumika Talware